



Vibe Sort

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Goal

The goal was to create a fast program for sorting 64-bit integers while staying within the 1 GB memory and 15-second runtime limits.

Work Process

I started by reading the assignment and then gave it to ChatGPT to better understand what was being asked and what sorting methods could work.

My first attempt was using Codex to generate a complete and optimized solution. It produced a ZIP file with the code, but the solution was too complicated for me to properly understand or explain. Because of this, I decided not to use it.

I then went back to ChatGPT and asked for a much simpler and more readable solution that only included what was necessary for the assignment. This produced a single Java file using radix sort, along with a README explaining the code.

I had some problems getting the program to run at first, which I troubleshooted with ChatGPT. After fixing these, I was able to compile and run the program.

Final Solution

The final program uses radix sort to sort Java long values. It processes the 64-bit numbers in smaller sections rather than comparing every number directly.

The program tests random, sorted, reverse sorted and partially sorted arrays. It also measures the sorting time for different array sizes to demonstrate its performance and scaling.

Example of an output from the program:

```
Random input scaling test
n,time_seconds
100000,0.003795
500000,0.060789
1000000,0.059426
2000000,0.099249
5000000,0.221907
10000000,0.464678
20000000,1.207299

Input type test with n = 5000000
Sorted:      0.236646 seconds
Partially sorted: 0.218718 seconds
Random:      0.211342 seconds
Reverse:     0.175757 seconds
```

Use of AI

I used ChatGPT through the ChatGPT website to understand the assignment, generate the code, troubleshoot errors and help with documentation. I also tried Codex for the first implementation but decided against using its solution because I did not understand the generated code well enough.

Conclusion

In conclusion, I had a very hard time understanding the assignment. I did not understand what was supposed to be delivered, what the final goal with the assignment was and just understood the assignment. I can understand the code that ChatGPT gave me, but I do not understand